

EMPLOYEE ATTRITION PREDICTION — SUMMARY REPORT

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College: Stella Maris College, Chennai

Program: BCA 3rd Year

DATASET: IBM HR Analytics — 1,470 employees, 35 features

ATTRITION RATE: ~16% (imbalanced dataset)

MODELS TRAINED: Logistic Regression, Random Forest, Gradient Boosting

BEST MODEL: Logistic Regression

KEY FINDINGS FOR HR DIRECTOR:

- 1 in 6 employees is likely to leave the company
- Sales Representatives have the highest exit rate (~40%)
- Employees earning lower salaries leave more frequently
- Overtime work is one of the strongest warning signs of resignation
- Most exits happen within the first 2-3 years of joining

RECOMMENDATIONS:

1. Create a structured retention program for new employees (0-3 years)
2. Monitor and reduce excessive overtime across all departments
3. Conduct salary reviews for Sales and HR departments urgently

LIMITATION: Model should support — not replace — HR conversations.